Chapter 3.1 — Formula Sheet

Sampling Mechanics & Finite Population Selection

Companion quick-reference to your Notebook 3 syllabus and your *20 Worked Examples* set (Example labels 1–20 cross-referenced below)

Hypergeometric Sampling

PMF & Setup [Ex 1–4, 11]

Population N, K marked, draw n without replacement:

$$\mathbb{P}(X=k)=\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \qquad \max(0,n-N+K)\le k\le \min(n,K)$$

Multivariate generalization (c categories, $\sum K_i = N$, $\sum n_i = n$):

$$\mathbb{P}=\frac{\prod_i \binom{K_i}{n_i}}{\binom{N}{n}}$$

Duality: Hypergeometric(N,K,n) $\overset{d}{=}$Hypergeometric(N,n,K) — always relabel so the *smaller* of K,n plays the "choose from" role to minimize arithmetic.

Mean, Variance & the FPC [Ex 5–8]

$$\mathbb{E}[X]=n\frac{K}{N}, \qquad \mathrm{Var}(X)=n\cdot\frac{K}{N}\cdot\frac{N-K}{N}\cdot\underbrace{\frac{N-n}{N-1}}_{\text{FPC}}$$

Derivation route: indicator $I_j=\mathbb{1}[\text{item } j \text{ marked \& drawn}]$, $\mathbb{E}[I_j]=n/N$; variance needs $\mathrm{Cov}(I_j,I_k)=-\frac{n(N-n)}{N^2(N-1)}$ for $j\neq k$ (negative dependence — drawing one removes it from the pool). **Rule of thumb:** $n/N<5\%\Rightarrow$ FPC≈ 1, Binomial is safe.

Approximations [Ex 9–10]

Binomial (valid when n/N small, $p=K/N$):

$$\mathbb{P}(X=k)\approx\binom{n}{k}p^k(1-p)^{n-k}$$

Normal (large n, continuity-corrected), using the exact Hypergeometric μ,σ^2 *including the FPC*:

$$\mathbb{P}(X\le x)\approx\Phi\left(\frac{x+0.5-\mu}{\sigma}\right)$$

Urn Models & Recapture

Pólya Urn Schemes [Ex 12–16]

Start (r,b), draw-and-reinforce by c of the same color each step. **Exchangeability:** $\mathbb{P}$(sequence with k of one color) depends only on k, not order:

$$\mathbb{P}=\frac{r^{(k)}\,b^{(n-k)}}{(r+b)^{(n)}} \quad \text{(rising factorials, } c=1\text{)}$$

Martingale: $M_n=\frac{R_n}{R_n+B_n}$ satisfies $\mathbb{E}[M_{n+1}\mid\mathcal{F}_n]=M_n\Rightarrow$ constant mean for *every* n.
Beta limit: $M_\infty\sim\mathrm{Beta}(r,b)$, so $\mathbb{E}[M_n]=\mathbb{E}[M_\infty]=\frac{r}{r+b}$ for all n — the discrete/continuous bridge to Bayesian conjugacy. Reinforcement strength c tunes momentum: $c=0$ is i.i.d. (no memory), $c\to\infty$ is near-instant lock-in.

Capture–Recapture [Ex 17–20]

Tag n_1, resample n_2, observe m_2 overlap. **Lincoln–Petersen (method of moments *and* Hypergeometric MLE):**

$$\hat{N}=\frac{n_1n_2}{m_2}$$

MLE derivation route: likelihood ratio $L(N)/L(N-1)=\frac{(N-n_1)(N-n_2)}{N(N-n_1-n_2+m_2)}$, crosses 1 exactly at $N=n_1n_2/m_2$.
Chapman's bias-corrected estimator (small-sample safe, avoids the $1/m_2$ blow-up):

$$\hat{N}_C=\frac{(n_1+1)(n_2+1)}{m_2+1}-1$$

Bias direction: $\mathbb{E}[1/m_2]>1/\mathbb{E}[m_2]$ by Jensen (Ch. 2.2) on the convex map $x\mapsto 1/x$, so raw $\hat{N}$ over-estimates on average.

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*One-tool-tree reminder: Hypergeometric is "Binomial that remembers what it already drew" (negative dependence, FPC shrinks variance); Pólya urns run that dependence the other direction (positive reinforcement, FPC-free, variance-*growing *lock-in toward a Beta limit); capture-recapture inverts the Hypergeometric model itself, trading a known N for an estimated one. Same finite-population machinery, three directions.*